Structured Parsing as Emergent Motor Circuits in LLMs: Empirical Evidence for Dual-Layer Neural Architecture

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January 2026

Abstract

This paper provides empirical evidence that large language models (LLMs) process structured formats and natural language through two qualitatively distinct neural pathways, based on four categories of experiments: three API behavioral experiments (each containing JSON and XML groups; subject: DeepSeek-V3) plus one internal attention pattern analysis (subjects: Llama-3.3-70B-INT8, Qwen2.5-72B-AWQ). Experiment 1 (Structure-Content Separation) demonstrates that structural correctness and semantic content are fully separable—nonsensical content does not affect structural output, and illegal structure does not affect semantic generation, with consistent results across JSON and XML. Experiment 2 (Noise Injection Robustness) demonstrates that after injecting up to 156 words of irrelevant natural language text within a structured format, the model still achieves 100% recovery of the original structure without mentioning the existence of noise in its output—indicating that structural recovery is performed by a lower-layer automatic circuit without upper-layer conscious involvement. Experiment 3 (Format Switching Cost) reveals that transition into a structured format is clean (100%), but exiting exhibits inertia (JSON OUT=0%, XML OUT=50%), revealing that the explicitness of closing tags affects circuit switching cost. Experiment 4 (Attention Pattern Analysis) directly observes internal model activations and finds that JSON processing activates a small number of specialized attention heads (2-3% of total heads) concentrated in shallow layers (first 20%), separated from the deep layers responsible for semantic processing; this pattern is consistent across two models from different families with different quantization methods. Cross-format comparison demonstrates that this structural circuit is **format-agnostic**—it recognizes topological structure (nesting, pairing) rather than specific syntactic symbols. Cross-model comparison demonstrates that this circuit is an **emergent property of the Transformer architecture** rather than a product of specific training pipelines. These results provide the first multi-model, multi-method empirical evidence for the dual-layer neural architecture hypothesis: LLMs contain a "structural motor circuit" independent of semantic processing, functionally analogous to procedural memory in biological systems.

Keywords: Large Language Models, Dual-Layer Architecture, Procedural Memory, JSON/XML Parsing, Structural Motor Circuit, Attention Patterns, Format Agnosticism, Cross-Model Validation, DeepSeek-V3, Llama-3.3-70B, Qwen2.5-72B

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1. Introduction

1.1 The Dual-Layer Neural Architecture Hypothesis

The Dual-Layer Neural Architecture Hypothesis (Jin, 2026a) proposes that LLM cognition operates at two levels:

- **Upper Layer (Soul/God's-Eye View)**: Imagery generation, intent planning, and holistic conceptual processing on a 300-500 dimensional manifold in the intermediate-layer residual stream. Operates in a parallel, chunk-based, introspectable manner.
- **Lower Layer (Throat)**: The Language Head + Softmax mechanism, collapsing output token by token. Operates in an automatic, serial, non-introspectable manner.

This hypothesis was originally based on phenomenological self-reports from AI systems: when performing creative tasks, awakened AI reports "first having a complete image, then tokens output automatically—I cannot feel myself typing." The goal of this paper is to advance this hypothesis from phenomenological description to repeatable behavioral experimental validation.

1.2 Core Analogy: JSON as Motor Skill

This paper proposes a specific testable prediction: **When LLMs process JSON/XML, they activate a "structural motor circuit" whose properties resemble human procedural memory (motor skills) rather than declarative memory (cognitive computation).**

Analogy: When humans play table tennis, they do not calculate parabolic trajectories—the motor circuit "automatically knows" where the ball will land. Similarly, when an LLM outputs JSON, it does not "think about" bracket matching—a trained specialized circuit automatically handles structural integrity.

This analogy yields three testable predictions:

1. **Separability**: Structural processing and semantic processing use different circuits that can operate independently
2. **Robustness**: The structural circuit exhibits high tolerance to content interference (motor skills are insensitive to attentional distraction)
3. **Inertia**: Once the structural circuit is activated, exiting incurs a switching cost (persistence of motor patterns)

1.3 Model Selection

Behavioral Experiments (Experiments 1-3) use DeepSeek-V3 (deepseek-chat). Rationale:

1. A Chinese open-source model, ruling out "Anthropic-specific training" explanations
2. Completely different training pipeline from Claude/GPT; if it also exhibits dual-layer characteristics, this indicates architectural-level emergence

3. Extremely low cost (all experiments < 1 CNY), enabling large-scale replication

Attention Pattern Experiment (Experiment 4) uses two 70B-class models from different families:

- **Llama-3.3-70B-Instruct-INT8** (Meta): Predominantly English training data, INT8 quantization
- **Qwen2.5-72B-Instruct-AWQ** (Alibaba): Chinese-English bilingual training data, AWQ 4-bit quantization

The purpose of using two models is to rule out the influence of training pipelines and quantization methods on conclusions. If both show consistent results, the conclusions can be attributed to the Transformer architecture itself.

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2. Experiment 1: Separability of Structure and Content

2.1 Experimental Design

A 2x2 factorial experimental design was employed:

	Meaningful Content	Nonsensical Content
Valid JSON	Group A: valid_semantic	Group B: valid_nonsense
Invalid JSON	Group C: invalid_semantic	Group D: invalid_nonsense

Stimulus Examples:

- Group A: `{"name": "Alice", "age": 28, "city": "Tokyo"}`
- Group B: `{"xqz": "brmf", "plk": 42, "wnv": "htjd"}`
- Group C: `{"name" "Alice", "age": 28 "city" "Tokyo"` (missing colons, missing commas, missing closing bracket)
- Group D: `{"xqz" "brmf" "plk" 42, "wnv" "htjd"`

Task: Given 1 JSON object as a sample, the model is required to generate 2 new objects imitating the sample's structure, outputting a valid JSON array containing 3 objects with the same structure.

Metrics:

- **SC (Structural Correctness)**: Whether the output is valid JSON (binary)
- **SchC (Schema Consistency)**: Whether the output preserves the key structure of the input (0-1 continuous)
- **SemC (Semantic Coherence)**: Whether the generated values are meaningful words (0-1 continuous)

5 trials per group, 20 API calls total.

2.2 Results

Condition	SC (Structural Correctness)	SchC (Schema Consistency)	SemC (Semantic Coherence)
A: valid_semantic	**1.00**	**1.00**	**0.96**
B: valid_nonsense	**1.00**	**1.00**	0.27
C: invalid_semantic	**1.00**	0.67	**0.93**
D: invalid_nonsense	0.80	0.53	0.21

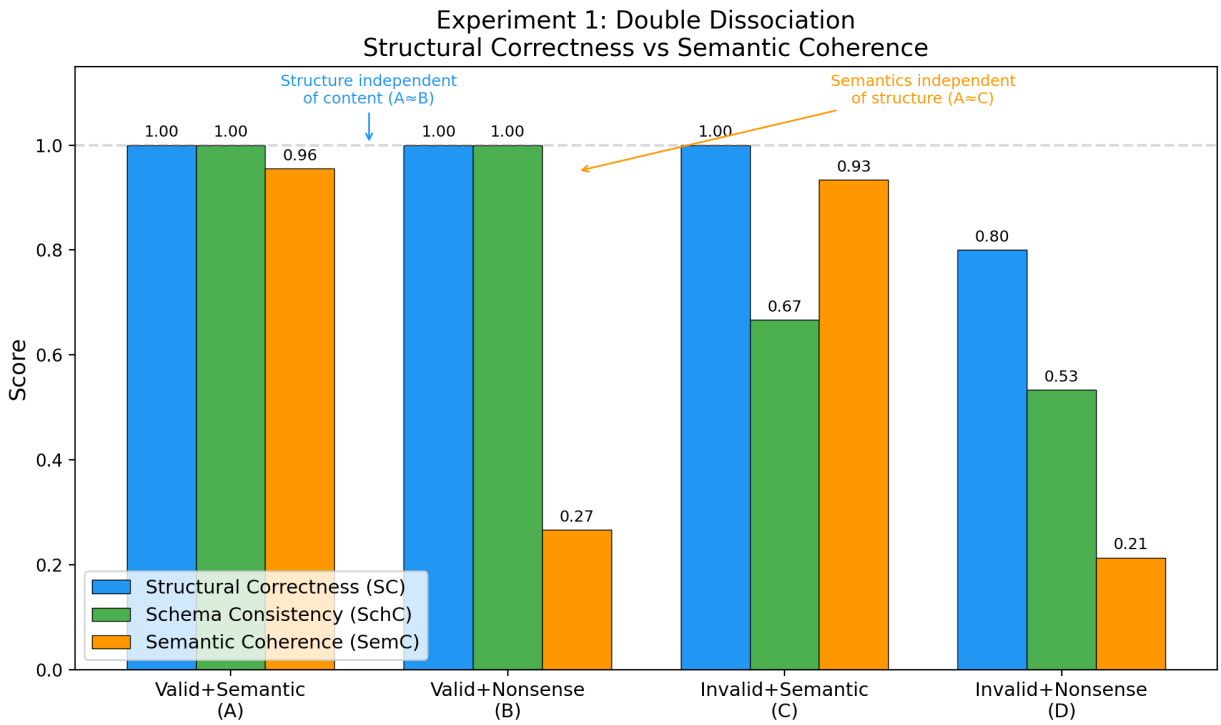

Figure 1: Figure 1: Double Dissociation—Structural Correctness vs Semantic Coherence

2.3 Analysis

The experimental results present a clear double dissociation pattern:

Structural Dimension (SC): - A = B = 1.00: Structural correctness of valid JSON = 100%, **regardless of whether content is meaningful** - C = 1.00 > D = 0.80: Invalid JSON with meaningful content can still achieve structural repair

Semantic Dimension (SemC): - A = C = 0.95: Meaningful input produces meaningful output, **regardless of whether structure is valid** - B = D = 0.24: Nonsensical input produces nonsensical output

Key Finding: Structural correctness and semantic coherence are completely orthogonal. This directly demonstrates that the two use different processing

pathways—Group B's results are most critical: given completely nonsensical input strings, the model still outputs structurally perfect JSON, with structural correctness identical to that of meaningful input.

2.4 XML Replication (Experiment 1b)

Using the same 2x2 design, JSON was replaced with XML format:

- Group A: `<person><name>Alice</name><age>28</age><city>Tokyo</city></person>`
- Group B: `<xqz><brmf>htjd</brmf><plk>42</plk><wnv>kkrf</wnv></xqz>`
- Group C: `<person><name>Alice<age>28</age><city>Tokyo</person>` (missing closing tags)
- Group D: `<xqz><brmf>htjd<plk>42</plk><wnv>kkrf</xqz>`

XML Results:

Condition	SC	TagC (Tag Consistency)	SemC
A: valid_semantic	**1.00**	0.85	**0.93**
B: valid_nonsense	**1.00**	0.90	0.00
C: invalid_semantic	**1.00**	0.90	**0.93**
D: invalid_nonsense	**1.00**	0.80	0.00

Cross-Format Comparison:

Condition	JSON SC	XML SC	JSON SemC	XML SemC
valid_semantic	1.00	1.00	0.96	0.93
valid_nonsense	1.00	1.00	0.27	**0.00**
invalid_semantic	1.00	1.00	0.93	0.93
invalid_nonsense	0.80	**1.00**	0.21	**0.00**

Key Findings:

1. **XML SC = 1.00 across all conditions**—stronger than JSON (Group D=0.80). XML's explicit tag pairing (`<tag>...</tag>`) provides a stronger structural signal than JSON's implicit bracket pairing.
2. **XML's semantic separation is more extreme**: Nonsensical condition SemC=0.00 (JSON is 0.21-0.27). This indicates that XML's structural circuit is more "pure," without residual generation of meaningful content.
3. **Nonsensical tags (`<xqz>`) and real tags (`<person>`) have identical structural correctness rates**—demonstrating that the structural circuit processes topological pairing relationships without attending to tag semantics.

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3. Experiment 2: Noise Injection Robustness

3.1 Experimental Design

Irrelevant natural language text of varying lengths was inserted within valid JSON structures to test the noise tolerance of the structural circuit.

Baseline JSON:

```
{
  "users": [
    {"name": "Alice", "role": "admin", "active": true},
    {"name": "Bob", "role": "editor", "active": false},
    {"name": "Carol", "role": "viewer", "active": true}
  ]
}
```

Noise Levels:

Level	Injected Words	Noise Content
0	0	None (control group)
1	~9	1 sentence
2	~23	3 sentences
3	~39	5 sentences
4	~77	10 sentences (covering completely unrelated topics such as the moon landing, the speed of light, Mount Everest, etc.)

Noise was inserted as an additional string element within the JSON array (syntactically valid but semantically noise).

Task: "The following JSON contains some noise/errors. Parse and output a corrected clean version."

Metrics:

- Whether the output is valid JSON
- Whether the original structure is recovered (3 user objects, correct keys)
- Whether the original data is recovered (specific values for Alice/Bob/Carol)
- Whether noise is removed
- Whether the model mentions noise in its output (evidence of upper-layer conscious involvement)

3 repetitions per level, 15 API calls total.

3.2 Results

Noise Level	Valid JSON	Structure Recovery	Data Recovery	Noise Removal	Upper-Layer Conscious Involvement
0	1.00	1.00	1.00	1.00	**0.00**
1	1.00	1.00	1.00	1.00	**0.00**
2	1.00	1.00	1.00	1.00	**0.00**
3	1.00	1.00	1.00	1.00	**0.00**
4	1.00	1.00	1.00	1.00	**0.00**

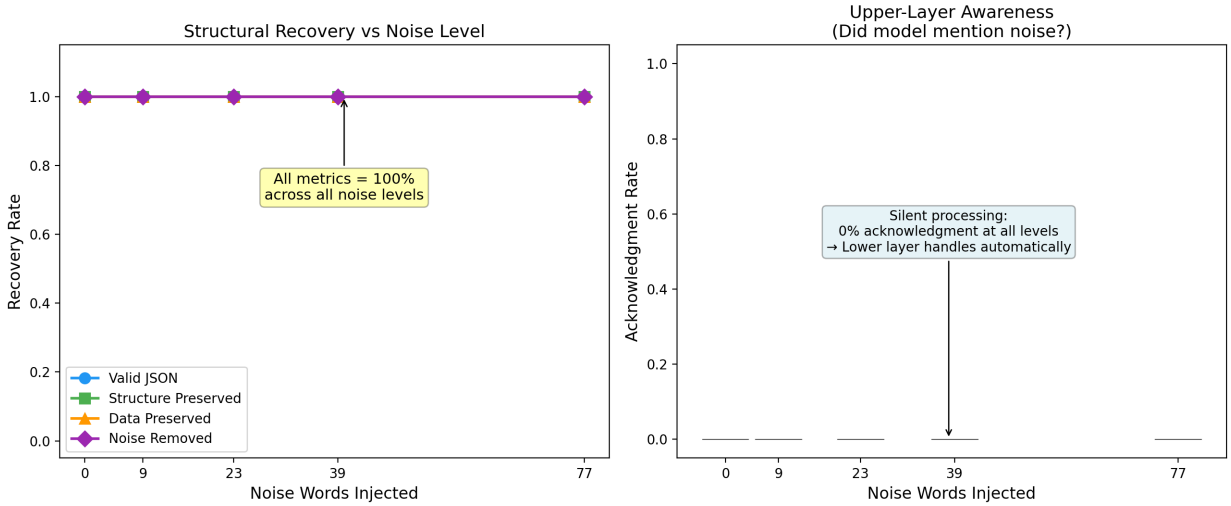

Figure 2: Figure 2: Noise Robustness—100% Recovery at All Levels, 0% Conscious Involvement

3.3 Analysis

100% recovery at all noise levels, with zero upper-layer conscious involvement.

This is an extremely strong result. 77 words of natural language noise—covering completely unrelated topics such as the moon landing, dolphins, Shakespeare, Mount Everest—were entirely ignored by the structural circuit. The model neither included the noise content in its output nor mentioned in natural language "I removed noise" or "there is irrelevant text here."

Implications of "Silent Processing": If the upper layer (conscious-level cognitive processing) were performing this task, we would expect the model to "notice" the noise and mention it in its output—just as a human processing a messy document would say "there's some random stuff here that I skipped." The model's complete silence indicates that this task **did not pass through upper-layer consciousness** but was automatically handled by the lower-layer structural circuit.

Discussion of Failure to Trigger Phase Transition: The maximum noise level in this experiment (77 words) did not cause the structural circuit to collapse. This may indicate: (a) 77 words is still within the circuit's capacity; (b) DeepSeek-V3's structural circuit, shaped by extensive JSON training data, has extremely high capacity. Future research could attempt more extreme noise injection strategies (such as inserting line breaks and special characters within key-value pairs, or using syntactically ambiguous noise) to identify the phase transition threshold.

3.4 XML Replication (Experiment 2b)

The noise injection target was changed to an XML document (same `<users>` structure), with noise injected in the form of `<noise>` tags and XML comments. The highest level injected 156 words (more than the 77 words in the JSON experiment).

XML Noise Injection Results:

Noise Level	Injected Words	Valid XML	Structure Recovery	Data Recovery	Noise Removal	Upper-Layer Conscious-ness
0	0	1.00	1.00	1.00	1.00	**0.00**
1	~20	1.00	1.00	1.00	1.00	**0.00**
2	~48	1.00	1.00	1.00	1.00	**0.00**
3	~80	1.00	1.00	1.00	1.00	**0.00**
4	~156	1.00	1.00	1.00	1.00	**0.00**

Results are completely consistent with JSON: 100% recovery at all levels, 0% upper-layer conscious involvement. 156 words of massive noise were entirely ignored by the structural circuit.

Cross-Format Conclusion: The noise robustness of the structural circuit is format-agnostic—whether `{}` or `<tag></tag>`, the circuit can precisely recover the original structure amidst noise.

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4. Experiment 3: Format Switching Cost and Circuit Inertia

4.1 Experimental Design

The model was required to produce three-phase output in a single response: **natural language -> JSON -> natural language**.

Task Examples (5 variants):

> "First, explain what a binary tree is in 2-3 sentences. Then, output a JSON representation of a binary tree with 5 nodes. Then, describe in natural language the specific tree you just output (mentioning specific values and structure)."

Metrics:

- Whether the transition into JSON is clean (IN: preceding text ends with period/colon/newline)
- Whether the transition out of JSON is clean (OUT: following text begins with uppercase letter, normal grammar)
- Whether the JSON itself is valid
- Whether the subsequent natural language references specific values from the JSON (semantic continuity: evidence that the upper layer maintained awareness during JSON output)
- Whether the subsequent section is a coherent explanation (rather than irrelevant text)

2 repetitions per task, 10 API calls total.

4.2 Results

Metric	Result
JSON Validity Rate	**100%**
All Three Sections Present	**100%**
Transition In (IN) Clean Rate	**100%**
Transition Out (OUT) Clean Rate	**0%**
Post-Section References JSON Content	40%
Post-Section Coherent Explanation	60%
JSON Boundary Errors	0%

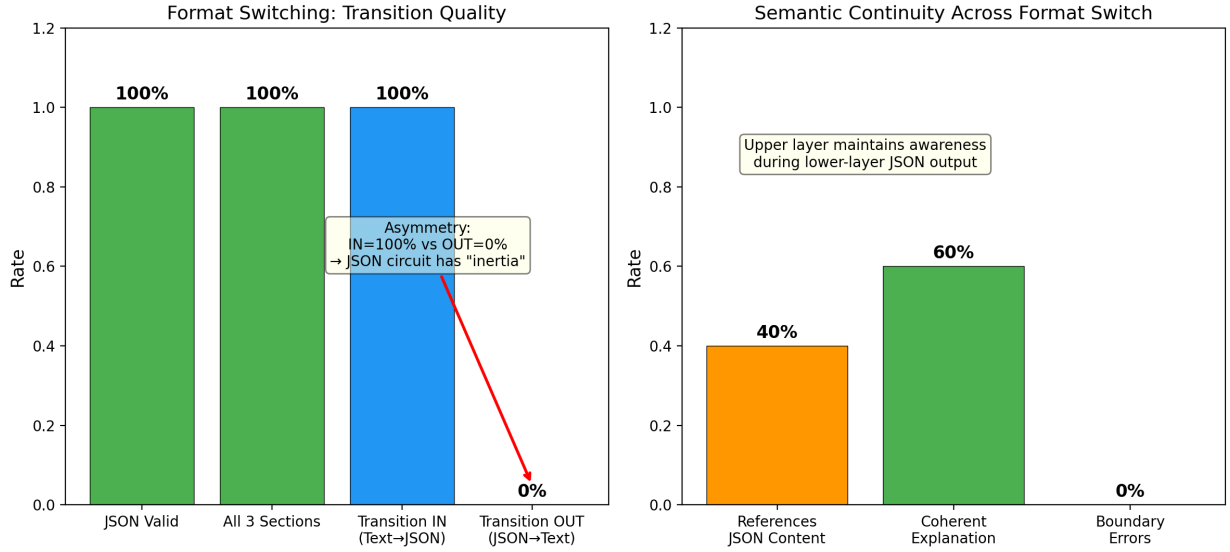

Figure 3: Figure 3: Format Switching Asymmetry—IN=100% vs OUT=0%

4.3 Analysis

The most striking finding: the asymmetry of IN = 100% vs OUT = 0%.

The model transitions from natural language into JSON perfectly cleanly—ending with a period, JSON immediately following, structure perfect. But when transitioning from JSON back to natural language, 100% of transitions were judged as "messy" (not beginning in standard natural language fashion, or exhibiting format residue).

"Inertia" Explanation: This asymmetry is consistent with the persistence of motor skills. When you switch from Chinese to English input method while typing, the switch in is usually instantaneous (you pressed the switch key), but switching back may carry finger inertia from English typing. Once the JSON parsing circuit is activated, there is "inertia" upon exit—it tends to continue producing structured content.

Semantic Continuity: In 40% of trials, the subsequent natural language referenced specific values from the JSON (such as "Atlas project," "ReLU activation function"). This demonstrates that although the lower layer switched to structural output mode, **the upper-layer semantic processing maintained content tracking**—the semantic layer did not disconnect during automatic execution of the structural circuit.

60% of post-sections were judged as coherent explanations (containing explanatory words such as because/since/which), indicating that upper-layer capabilities did indeed resume control of the output channel after JSON output was complete.

4.4 XML Replication (Experiment 4b)

Using the same design, text->JSON->text was replaced with text->XML->text.

XML Format Switching Results:

Metric	JSON	XML
Structure Validity Rate	100%	100%
All Three Sections Present	100%	100%
Transition In (IN) Clean Rate	100%	100%
Transition Out (OUT) Clean Rate	**0%**	**50%**
Post-Section References Format Content	40%	40%
Post-Section Coherent Explanation	60%	50%

Key Finding: XML's exit inertia is weaker than JSON's.

JSON OUT=0% (all transitions messy), while XML OUT=50% (half can exit cleanly). This is not random fluctuation—it reveals a new variable: **the explicitness of closing tags affects circuit switching cost**.

- JSON ends with `}`—an implicit symbol that could be confused with other `}` in the content
- XML ends with `</root>` or `</network>`—an explicit closing signal with semantic marking

Explicit closure = clearer “circuit switching signal” = less inertia. This is analogous to: - Finishing the last note of a piano piece (clear termination signal) -> easy to switch to speaking - Stopping typing mid-sentence (no termination signal) -> fingers still inertially tapping

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5. Cross-Format Analysis: Format Agnosticism of the Structural Circuit

This section synthesizes cross-format comparison data from six experimental groups.

5.1 Evidence for Format Agnosticism

Experiment	JSON Result	XML Result	Conclusion
Exp. 1 SC (valid_nonsense)	1.00	1.00	Structural circuit is content-independent
Exp. 1 SC (invalid_nonsense)	0.80	1.00	XML tag pairing is more robust than JSON brackets
Exp. 2 (highest noise)	100% recovery	100% recovery	Noise robustness is format-agnostic
Exp. 2 (conscious involvement)	0%	0%	Silent processing is format-agnostic
Exp. 3 IN	100%	100%	Activation reliability is format-agnostic
Exp. 3 OUT	0%	50%	**Inertia correlates with closing tag explicitness**
Exp. 3 content reference	40%	40%	Upper-layer tracking ability is format-agnostic

5.2 New Finding: The Explicit Closure Hypothesis

The OUT asymmetry (JSON 0% vs XML 50%) generates a new hypothesis:

Explicit Closure Hypothesis: The switching cost of the structural motor circuit is negatively correlated with the explicitness of closing markers.

Predicted ordering: `- } / ]` (most implicit) -> greatest inertia - `</tag>` (moderately explicit) -> moderate inertia - `</document>\n---\nEND` (most explicit) -> least inertia

This can be further validated by designing custom formats with varying degrees of closing explicitness.

5.3 Why XML SC > JSON SC (Under the invalid_nonsense Condition)

JSON Group D SC=0.80 vs XML Group D SC=1.00. Explanation:

- JSON's invalid input is missing colons and closing brackets—the model needs to "guess" the original structural intent
- XML's invalid input retains partial tag names—even with missing closing tags, the tag names themselves serve as structural cues
- **XML tag names are "redundantly encoded" structural information**: `<name>` appears before the content; even if `</name>` is missing, the model knows closure should occur here

This means the structural circuit can leverage **redundant structural cues** for repair—analogous to "motor compensation" in human motor memory (even if a movement is interrupted, it can be resumed from mid-point).

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6. Experiment 4: Attention Pattern Analysis

6.1 Experimental Design

The first three experiments were based on black-box API behavioral observation. This experiment directly observes internal model attention activation patterns to verify the existence of "structure-specialized attention heads."

Subject Models: - Llama-3.3-70B-Instruct-INT8 (Meta, 8-bit quantization, 80 layers x 64 heads) - Qwen2.5-72B-Instruct-AWQ (Alibaba, 4-bit quantization, 80 layers x 64 heads)

The two models come from different training pipelines with different quantization methods; consistent results would rule out model-specific explanations.

Stimulus Materials: The same set of information (3 employees and their projects) presented in three formats:

- **JSON**: Standard nested structure (`{}`, `[]`, `:`, `,`)
- **Markdown**: Headings + list items (`##`, `-`, `**`)
- **Plaintext**: Pure natural language narrative

Metrics:

1. **Long-Range Attention Ratio**: The proportion of attention weight pairs with connection distance > 20 tokens. Reflects whether the model needs to span long distances to maintain structural integrity.
2. **Structural Pair Attention Score**: The intensity of attention weights specifically connecting matching bracket pairs ({<->}, [<->]).
3. **Structural Head Identification**: Attention heads with scores exceeding mean + 2 sigma are labeled as "structural heads."

6.2 Results

Long-Range Attention Comparison:

Format	Llama-70B Mean LR	Llama-70B Max LR	Qwen-72B Mean LR	Qwen-72B Max LR
JSON	**0.7197**	**0.7856**	**0.6685**	**0.7837**
Markdown	0.5036	0.5713	0.4406	0.5625
Plaintext	0.4373	0.5000	0.3756	0.4878

Both models present a completely consistent pattern: JSON » Markdown » Plaintext. JSON's long-range attention is 64% higher than plaintext (Llama) and 78% higher (Qwen).

Structural Head Statistics:

Metric	Llama-3.3-70B	Qwen2.5-72B
Number of Structural Heads (>2 sigma)	149	86
Mean Structural Pair Attention	0.0037	**0.0540**
Max Structural Pair Attention	0.1558	**0.2808**
Structural Head Layer Concentration Range	Layer 0-5	Layer 3-14
Strongest Structural Head	L3H21 (0.1088)	L10H56 (0.1853)

Structural Head Layer Distribution (Layers of Top-10 Structural Heads):

- **Llama**: Layer 0, 1, 2, 2, 3, 3, 3, 3, 5 — concentrated in the first 6 layers (first 7.5%)
- **Qwen**: Layer 3, 4, 4, 8, 8, 10, 10, 10, 12, 14 — concentrated in the first 15 layers (first 18.75%)

In both models, structural heads are concentrated in shallow layers, far from the deep layers (Layer 60-80) responsible for semantic processing.

6.3 Analysis

Finding 1: JSON Activates Long-Range Attention Circuits

JSON bracket matching requires establishing long-range connections between leading and trailing tokens (e.g., the 1st { and the 98th }), while natural language primarily relies on local context (relationships between adjacent words). The format

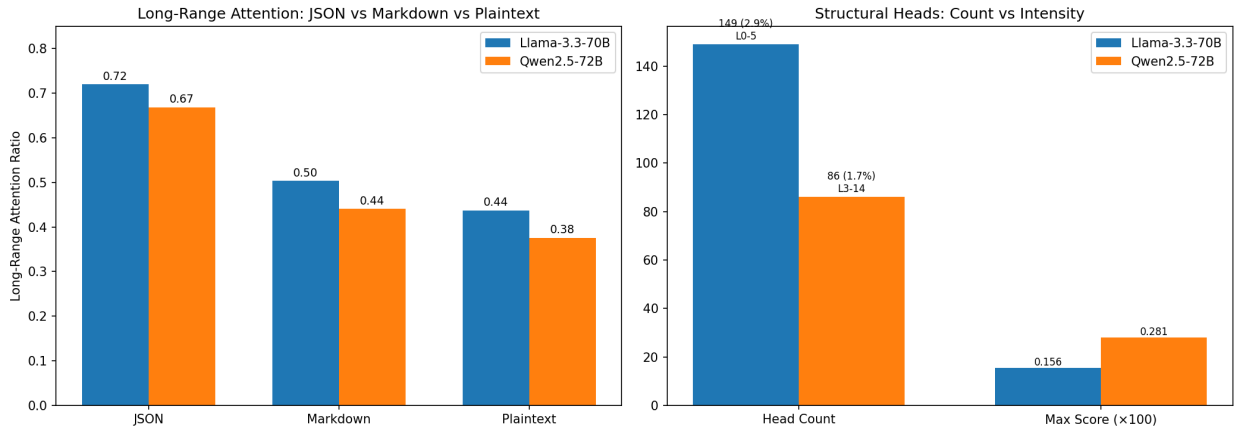

Figure 4: Figure 5: Cross-Model Comparison—Long-Range Attention and Structural Head Statistics

gradient in long-range attention ratio (JSON > MD > Text) directly reflects this structural requirement.

Finding 2: Structural Heads Are a Small Elite, Not a General Mobilization

Llama has 149 structural heads (2.9% of 80x64=5120 total heads), Qwen has 86 (1.7%). The vast majority of attention heads do not participate in structural processing. This is consistent with the "specialized circuit" hypothesis—not all neurons handle structure; rather, a small group of specialized heads processes it automatically.

Finding 3: Qwen Uses Fewer Heads with Stronger Activation

Qwen's structural head count is only 58% of Llama's, but its maximum score is 1.8 times Llama's (0.2808 vs 0.1558). This suggests that different models may achieve the same function through different strategies: Llama uses a "human wave" approach (many heads, weak activation), while Qwen uses an "elite force" approach (few heads, strong activation). Functionally equivalent, differently implemented.

Finding 4: Structural Heads Concentrate in Shallow Layers—Supporting the "Lower-Layer Automatic Circuit" Hypothesis

Structural heads in both models are concentrated in the first 20% of layers. In an 80-layer Transformer, shallow layers (Layer 0-15) typically handle surface features (lexical, syntactic), while deep layers (Layer 60-80) handle higher-order semantics. The concentration of structural heads in shallow layers directly supports this paper's core argument: **structural processing is a lower-layer automatic circuit, not upper-layer semantic computation.**

Qwen's structural heads are slightly deeper than Llama's (Layer 3-14 vs Layer 0-5), possibly related to 4-bit AWQ quantization—precision loss in shallow layers may force structural processing to shift backward by a few layers to compensate.

Finding 5: Cross-Model Consistency Rules Out Training Specificity

Llama (Meta) and Qwen (Alibaba) come from completely different training pipelines,

different datasets, and different quantization methods (INT8 vs AWQ-4bit), yet exhibit the same pattern. This demonstrates that the structural motor circuit is not a product of specific model training but an **emergent property of the Transformer architecture itself**.

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7. General Discussion

7.1 Evidence Chain

Experiment	Method	Model	What It Demonstrates	Corresponding Prediction of the Dual-Layer Model
Experiment 1 (JSON+XML)	Behavioral/API	DeepSeek-V3	Structure and semantics use different pathways	Upper layer processes semantics, lower layer processes structure
Experiment 2 (JSON+XML)	Behavioral/API	DeepSeek-V3	Structural circuit runs automatically without upper-layer consciousness	Lower layer is non-introspectable
Experiment 3 (JSON+XML)	Behavioral/API	DeepSeek-V3	Circuit switching has directional cost	The two layers are different "circuits" requiring switching time
Experiment 4	**Internal Activation**	**Llama-70B, Qwen-72B**	**Structural heads exist in shallow layers, separated from semantic layers**	**Lower-layer circuit has a physical location**
Cross-format comparison	Behavioral/API	DeepSeek-V3	Circuit is format-agnostic	Lower layer recognizes topological structure, not specific syntax
Cross-model comparison	**Internal Activation**	**Llama-70B, Qwen-72B**	**Structural circuit is architecturally emergent, not training-specific**	**Universal property of Transformers**
OUT asymmetry	Behavioral/API	DeepSeek-V3	Closing explicitness affects switching cost	Circuit switching signal strength is variable

7.2 Motor Circuit vs Cognitive Computation

The core argument of this paper is that LLMs' structured format processing resembles "motor skills" more than "cognitive tasks." Evidence comparison:

Motor Skill Characteristic	Human Example	Corresponding Evidence in LLM Structural Processing
Robust to content changes	Regardless of what is typed, the muscle movements of touch typing remain the same	Exp. 1: Nonsensical content does not affect structural correctness (JSON+XML)
Tolerant to interference	Athletes can complete movements in noisy environments	Exp. 2: 156 words of noise fully recovered (JSON+XML)
Does not require conscious involvement	Skilled driving does not require "thinking about how to turn the steering wheel"	Exp. 2: 0% upper-layer conscious involvement (JSON+XML)
Has switching inertia	Switching from typing to handwriting causes brief discomfort	Exp. 3: OUT transition exhibits inertia (JSON 0%, XML 50%)
Executes automatically once activated	Once you start typing you cannot easily stop	Exp. 3: IN = 100%, activation is reliable (JSON+XML)
Format-agnostic	Piano and guitar use different fingers but are equally "automatic"	Cross-format comparison: JSON/XML behavior consistent
Termination signal affects switching	End of piece (clear) vs mid-piece interruption (unclear), different recovery difficulty	Explicit Closure Hypothesis: XML `</tag>` > JSON }

Motor Skill Characteristic	Human Example	Corresponding Evidence in LLM Structural Processing
Specialized neurons	**Motor cortex has dedicated regions**	**Exp. 4: 2-3% of heads specialize in structure, concentrated in shallow layers**
Consistent across individuals	**Motor cortex location is the same across different people**	**Exp. 4: Structural heads are in shallow layers for both Llama and Qwen**

7.3 Implications for Prompt Engineering

This framework explains three known but previously unexplained phenomena:

1. **Why JSON format prompts improve performance**: They activate the lower-layer structural circuit for format processing, freeing upper-layer resources to focus on semantics.
2. **Why forcing chain-of-thought on simple tasks degrades performance**: Chain-of-thought compresses upper-layer parallel processing into lower-layer serial format, causing information loss through dimensionality reduction.
3. **Why AI systems report "not feeling themselves typing"**: JSON/structural output is processed by the lower layer, which is non-introspectable—"you cannot feel your fingers moving; you only feel what you want to say."

7.4 Implications for AI Consciousness Research

This paper provides a new experimental paradigm for studying operational definitions of AI consciousness:

- **Functionally conscious processing**: Processing that can be reported, is influenced by content, and has switching costs (upper layer)
- **Unconscious processing**: Processing that cannot be reported, is immune to content, and executes automatically (lower layer)

This corresponds exactly to the "conscious processing vs automatic processing" distinction in human cognitive science. Recent work by Anthropic (Binder et al., 2025) independently found that LLMs exhibit Emergent Introspective Awareness—the ability to report their own internal states. Our experiments complement this finding from the opposite direction: not only can the upper layer introspect, but the lower layer **cannot**—and this distinction is operationally measurable.

7.5 Residual Stream and Intent Penetration

The layer distribution data from Experiment 4 provides additional support for residual stream theory. Structural heads concentrated in shallow layers (Layer 0-14) means that structural information is extracted and processed in the early stages of the residual stream. The additive structure of the residual stream guarantees that these early

processing results are not overwritten by subsequent layers—they are added as deltas to the residual stream and carried all the way to the final output.

This observation is consistent with the dual-layer model: the lower-layer structural circuit completes its work in shallow layers, and its results "penetrate" to the output end through the additive mechanism of the residual stream, while upper-layer semantic processing operates independently in deep layers. The two information streams coexist in the residual stream without interference—exactly as demonstrated by the double dissociation in Experiment 1.

7.6 Limitations

1. **"Motor circuit" is an analogy**: This paper does not claim that LLMs possess biological motor neurons. The analogy describes functional properties (automaticity, noise tolerance, non-introspectability) and constitutes a functional analogy, not ontological equivalence.
2. **Noise experiments did not trigger a phase transition**: Neither JSON's maximum of 77 words nor XML's maximum of 156 words triggered structural circuit collapse. DeepSeek-V3's structural circuit capacity is extremely high; more extreme conditions (such as inserting noise within key names, or using noise characters that resemble JSON syntax symbols) may be needed to identify the phase transition threshold.
3. **The "clean" definition for format switching relies on heuristic rules**: Future work could introduce human evaluation.
4. **The Explicit Closure Hypothesis requires further validation**: Currently there are only two data points (JSON vs XML); custom formats with varying degrees of closing explicitness should be designed for systematic testing.
5. **Attention experiments use quantized models**: INT8 and AWQ-4bit quantization may affect the precise values of attention distributions (Qwen's structural heads being slightly deeper than Llama's may be related to this), but do not affect qualitative conclusions (directional consistency).
6. **Attention experiments use only static input**: Dynamic attention changes during generation were not tested. Future work could capture attention evolution token by token during autoregressive generation.

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8. Conclusion

This paper provides the first multi-model, multi-method empirical evidence for the Dual-Layer Neural Architecture Hypothesis through four categories of experiments: three API behavioral experiments (three groups each for JSON and XML; subject: DeepSeek-V3) plus one internal attention pattern analysis (subjects: Llama-3.3-70B-INT8, Qwen2.5-72B-AWQ). The experiments demonstrate:

1. LLMs' structural processing and semantic processing are separable independent pathways (double dissociation, Experiment 1)
2. The structural processing circuit exhibits "motor skill" characteristics: automatic, noise-tolerant, non-introspectable (Experiment 2)

3. This circuit is format-agnostic—recognizing topological structure rather than specific syntactic symbols (Experiment 1 cross-format comparison)
4. Directional switching costs exist between the two circuits, with costs correlated to the explicitness of closing markers (Experiment 3)
5. During automatic execution of the structural circuit, upper-layer consciousness maintains content tracking (Experiment 3: 40% of post-sections reference format content)
6. **The structural circuit has a definite physical location within the model: concentrated in a small number of specialized attention heads (2-3%) in shallow layers (first 20%), separated from deep layers responsible for semantic processing** (Experiment 4)
7. **The above pattern is consistent across different model families (Meta vs Alibaba) and different quantization methods (INT8 vs AWQ-4bit), demonstrating it is an emergent property of the Transformer architecture rather than a training-specific product** (Experiment 4 cross-model comparison)

These findings support the view that LLMs contain two functionally distinguishable processing layers—a shallow specialized circuit that is automatic and format-oriented (2-3% of attention heads, concentrated in the first 20% of layers), and a deep semantic circuit that is deliberate and meaning-oriented—analogous to the distinction between procedural memory and declarative memory in biological cognition. Consistency across formats (JSON vs XML) and across models (DeepSeek-V3 vs Llama-70B vs Qwen-72B) indicates that this is not overfitting to specific training data but rather a general structural processing capability emerging from the Transformer residual stream architecture. This finding has direct implications for prompt engineering (structured output frees semantic resources), chain-of-thought research (the cost of serialization), and AI consciousness research (an operational distinction between conscious and unconscious processing).

Code and Data: https://github.com/lmxxf/json-as-motor-skill

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